

- Daphnia pulex*, 448t, 1021, 1199f, 1200
- Darkness
 flowering in long-night plants and, 859f, 860
 plant etiolation response to, 843f
- Dark responses, rod cell, 1121f
- D'Arrigo, Rosanne, 773f
- Darwin, Charles. *See also* Evolution
 on barnacles, 710
 Beagle voyage and field research of, 471, 472f–473f
 on coevolution of flower-pollinator mutualism, 825
 on earthworms, 704
 evidence supporting theory of, 476, 477f–482f, 483
 on grandeur of evolutionary process, 484
 historical context of life and ideas of, 469f–471f
 on island species, 483
 on lung evolution from swim bladders, 728
 on mystery of flowering plants, 477, 647, 825f
 on mystery of speciation, 506f, 507
 on natural selection, 266, 267f, 487
 On the Origin of Species by Means of Natural Selection by, 469, 473, 483–484, 487
 speciation theory of, 473, 474f–476f
 on species diversity of tropics, 1232
 study by, of phototropism in grass coleoptiles, 847f
 theory of descent with modification by, 14f–16f, 469–470, 473f–476f, 483–484
 timeline of work of, 469f
- Darwin, Francis, 847f
- Dasyatis americana*, 727f
- Data, **17**, 19f, 21–22. *See also* Scientific Skills Exercises
- Databases
 in estimating reproductive rates, 1194, 1195f
 genome-sequence, 444, 445f
- Dating, radiometric, **32**–33, 529, 530f
- dATP, 324
- Daughter cells, 234, 235f–236f, 237, 266f
- Day-neutral plants, **859f**
- db* gene, 918
- DDT pesticide, 158, 485, 518, 1276f
- Dead-leaf moth (*Oxytenis modestia*), 476f
- Dead Sea, 585
- Dead zone, 1242, 1275f
- Deamination, amino acid, 182
- Deaths
 demographics of, 1193t, 1194f–1195f
 in density-dependent population growth, 1203f–1205f
 in exponential population growth, 1196, 1197f
 in human population dynamics, 1208–1209
 in population dynamics, 1192, 1256f
- Death signals, apoptosis, 230, 231f. *See also* Apoptosis
- Decapods, 709f, 710
- Decay curves, radioactive isotope, 33
- December solstice, 1167f
- Deciduous forest, nutrient cycling in, 1252f
- Decomposers, **188**, **586**, **1240f**
 in energy flow and chemical recycling, 9f, 1240f, 1241
 fungi as, 654–655, 658f, 661f, 663, 665, 667
 lichens as, 669
 prokaryotic, 586
- Decomposition, 1240f, 1241, 1248f–1251f, 1259
- Deconvolution microscopy, 95f, 96
- Deductive reasoning, **18**, 617
- Deep-sea hydrothermal vents, 527f, 1182f
- Deer, 871, 1149, 1271
- Deer mice, 498f, 892
- DEET insect repellent, 1123
- De-etiolation (greening), **843f**–844f, 845
- Defensive adaptations, 28f, 43, 1217, 1218f–1219f, 1220, 1237
- Defensive proteins, 76f
- Deficiencies, plant, 810, 811f, 818
- Deforestation
 climate effects of, 1169f
 experimental, and nutrient cycling, 1252f
 as human community disturbance, 1231
 loss of species from, 1260f, 1263
 rising atmospheric carbon dioxide levels from, 1278
 of tropical rain forests, 651f, 652, 1263
- Degradation, protein, 379
- Dehydration
 animal, 980
 plant, 203, 204f–206f, 536
- Dehydration reactions, **67f**
- Dehydrogenases, 167f, 168
- Dejac, Lynn, 24f
- Delayed reproduction, 1208
- Deletions (mutations), 358f, **359**
- Deletions, chromosome, **307**, 308f, 309
- Dementia, 1103, 1104f
- Demographics, population, 1193t, 1194f–1195f
- Demographic transition, **1208**
- Demography, **1193**. *See also* Demographics, population
- Denaturation, **82f**, 83
- Dendrites, 879f, **1068f**, 1078, 1110f
- Dendritic cells, **955**
- Dendrobates pumilio*, 524
- Dendrochronology, 773f
- Dendroctonus ponderosae*, 1244f, 1245, 1280f
- Dengue fever, 412
- Denitrifying bacteria, 1251f
- Density, population, **1191f**–1192f, 1193, 1202, 1203f, 1256f
- Density-dependent inhibition, **247**, 248f
- Density-dependent population change, 1202, **1203f**–1206f, 1207, 1213
- Density-independent population change, **1203**
- Dentition, 530, 531f, 532, 911f, 912
- Deoxyribonucleic acid. *See* DNA
- Deoxyribose, **85f**, 317f, 324
- Dependent assortment hypothesis, 275, 276f
- Dependent variables, **21**, 513
- Dephosphorylation, protein, 223
- Depolarization, **1072f**–1074f
 in fertilization, 1044f, 1045
- Depolymerization, protein, 240
- Depression, 1081, 1102–1103
- Derelle, Romain, 612f
- Derived characters, shared, 560, 561f
- Derived traits, plant, 620f–621f
- Dermal tissue system, plant, 758, **762**, 763f
- DES (diethylstilbestrol), 1015
- Descending limb, loop of Henle, 988f, 989
- Descent with modification theory, 14f–16f, 469–470, 473f–476f, 483–484. *See also* Evolution
- Desert animals, 980–981
- Desert iguana (*Dipsosaurus dorsalis*), 888
- Desert mice, 981
- Deserts, 798, 799f, 1168, 1169f, **1173f**
- Desert spring, 1239f
- Desiccation, 979
- Desmognathus ochrophaeus*, 513
- Desmosomes, 120f
- Desynchronization, 858–859
- Determinate cleavage, **681**, 682f
- Determinate growth, **766f**–767f
- Determination, **383**, 384f, **1057**
- Detoxification, 104–105, 112, 809
- Detritus, **1177**, **1240f**, 1241
- Deuteromycetes, **659**
- Deuterostome development, **681**, 682f
- Deuterostomes, **683f**, 689f, 713, 714f–715f, 719f.
 See also Chordates; Echinoderms
- Development, **775**
 of brain, 722f, 1092f, 1099–1100
 as cell division function, 235f
 embryonic. *See* Embryonic development
 in human life cycle, 257f, 258. *See also* Human
 embryonic development
 macroevolution of, 545f–546f, 547
 plant and animal, 894f. *See also* Animal
 development; Plant development
 postzygotic barriers and, 509f, 510
 as property of life, 3f
 sustainable. *See* Sustainable development
- Developmental biology, 1057
- Developmental genes. *See* Homeotic genes; *Hox* genes
- Developmental plasticity, 775f
- Developmental potential, cell fate and, 1060, 1061f
- Diabetes
 aquaporin mutations as causes of diabetes
 insipidus, 995f
 autoimmunity and, 971
 disruption of glucose homeostasis in, 916–917
 genetic engineering of insulin to treat, 436
 genetic markers for, 427
 mitochondrial mutations in, 311
 neonatal, insulin mutations and, 359
 stem cells for, 431
- Diabetes insipidus, 995f
- Diabetes mellitus, **916**–917
- Diadocedexis*, 482f
- Diacylglycerol (DAG), **225f**
- Diagnosis
 antibodies as tools in, 969f
 biotechnology in, 433, 434f, 435
- Diamond-back rattlesnake (*Crotalus atrox*), 1084
- Diaphragm in breathing, **945f**
- Diaphragm, birth control, 1038f
- Diapsids, **736**
- Diarrhea, 139, 224, 588, 598f, 911
- Diastole, **927f**
- Diastolic pressure, **930**, 931f
- Diatomaceous earth, 602
- Diatoms, 244f, 599f, **602f**, 617, 1170
- Diazepam, 1081
- Dicer-2, 954f
- Dicots, **649**
- Dictyostelium discoideum*, 613f
- Dideoxy DNA sequencing, 416
- Dideoxyribonucleotide (dideoxy) chain termination
 DNA sequencing, 443
- Didinium*, 617
- Diencephalon, 1093f
- Diets. *See also* Food
 adaptations of vertebrate digestive systems for, 911f–914f
 assessing nutritional needs in, 902f
 catabolism and human, 182f, 183
 deficiencies in, 901f, 902
 essential nutrients for, 899f, 900t–901t, 901f
 genetic variation in prey selection and, 1155, 1156f
 nonheritable variation and, 488f
 phenylketonuria and, 492
 typical and opportunistic, 901
- Differential centrifugation, 96f, 97
- Differential gene expression, **370**. *See also* Gene regulation
 cytoplasmic determinants and induction in, 382f, 383
 in eukaryotic gene regulation, 370f, 371
 in gene regulation, 365f
 in pattern formation of body plans, 384, 385f–387f
 in processes of embryonic development, 381f, 382. *See also* Embryonic development
 sequential regulation of, during cellular differentiation, 383, 384f
- Differential-interference microscopy, 95f
- Differential reproductive success, 266, 267f
- Differential speciation success, 548–549
- Differentiation, **1057**
- Differentiation, cell, **381**
 cytokinins in, 850
 as embryonic development process, 381f, 382.
 See also Embryonic development
 plant development and, 763, 775–776, 778f
 sequential gene regulation in, 383, 384f
 stem cells and, 430f, 431. *See also* Stem cells
- Diffusion, **132**, **922**
 body surface area and, 695f
 effects on water balance of osmosis as, 133f–135f
 extracellular, 768
 free energy change and, 148f
 interpreting scatter plots on glucose uptake as, 136
 as passive transport, 132, 133f
 in plant cells, 209f
 proteins and facilitated, 135f, 136
 of water across plant plasma membranes, 788, 789f–791f
 of water and minerals into root cells, 792
- Digestion, **904**
 in animal food processing, 904f–905f
 compartmentalized processing in, 898
 digestive compartments in, 902, 903f–905f
 digestive systems and, 875f
 extracellular, 904f–905f
 feeding mechanisms and, 902, 903f
 fungal, 655, 668
 hydrolysis in, 67
 intracellular, 904
 lysosomes in intracellular, 107f, 108
 regulation of animal, 915f
 sea star, 713, 714f